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IGNITION SYSTEM AND METHOD FOR MANUFACTURING THE SAME, GAS GENERATOR AND SAFETY AIRBAG

Abstract

An ignition system (**10**) for a gas generator comprises an ignition tube (**1**); a base (**2**) having a receptacle (**23**), wherein the ignition tube (**1**) is press-fitted into the receptacle (**23**) and extends out of the base (**2**) with its head (**11**), and the base (**2**) is configured as an integral injection-molded part, wherein a metal inlay (**4**) is injection-molded into a plastic material, and the inlay (**4**) provides a portion of the receptacle (**23**); and a container (**3**) having an opening (**33**), through which the base (**2**) together with the ignition tube (**1**) is press-fitted into the container (**3**), with the head (**11**) of the ignition tube (**1**) facing forward.

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Background/Summary

TECHNICAL FIELD

[0001] The present application relates to an ignition system for a gas generator, and a method for manufacturing the same, as well as a gas generator comprising the same, and a safety airbag comprising the gas generator.

BACKGROUND OF INVENTION

[0002] Safety airbags are widely used as safety protection devices for occupants in transportation means, especially in land transportation means, in particular in motor vehicles. A safety airbag may comprise a folded cushion and a gas generator. In a predetermined operation condition of a transportation means, for example, in a collision event of a motor vehicle, a gas generator can be activated to generate gas rapidly, wherein the generated gas is filled into a cushion to inflate and pop up the folded cushion, wherein an occupant is buffered, wherein the head or other body parts of the occupant is prevented from colliding with a hard or sharp object. The gas generator may also be referred to as an inflator. The safety airbag may be, for example, a driver's safety airbag mounted in a steering wheel, a co-driver's safety airbag mounted in an instrument panel, a curtain-type safety airbag or the like.

[0003] In the vehicle technology, an inflatable restraint system, such as a safety airbag, has been developed as a supplement to a conventional safety belt restraint system. Typically, when a safety airbag is activated, an occupant in a vehicle can be protected in a way that the inflated cushion with elasticity absorbs the physical impact caused by collision of the vehicle, wherein in the collision event, the cushion is inflated and enters a space between the occupant and an internal object or an internal surface of the vehicle, wherein the inflated cushion buffers the occupant, thereby reducing the possibility of injury to the occupant due to the unexpected contact of the occupant with the internal object or the internal surface of the vehicle.

[0004] A safety airbag may be a part of a safety protection system comprising a plurality of structural assemblies in a vehicle. Typically, these structural assemblies may relate to a cushion, a gas generator, a sensor and an electronic control unit. The sensor can acquire data related to an emergency state of the vehicle. The electronic control unit can process the data, and thus can determine whether a collision is about to occur or is occurring. With a trigger signal being input to the gas generator, the gas generator can be powered on and activated and thus can supply or generate gas.

[0005] A gas generator is known in practice, wherein a chamber surrounding an ignition tube is sealed with a rupture membrane. When the ignition tube is activated, a shock wave is generated by the ignition tube and breaks the rupture membrane. Typically, the rupture membrane needs to be mounted by welding, in order to hermetically close a container of the ignition system. Such a manufacturing process for the ignition system is costly.

[0006] In addition, a compression gas generator is known in practice, which may also be referred to as a “compression gas inflator”. Such a gas generator comprises a pressure container, in which a pressure chamber for containing the compression gas is defined. When an ignition tube of the gas generator is activated, the ignition tube can ignite a pyrotechnic charge in a pyrotechnic charge chamber to generate a high-temperature and high-pressure gas instantaneously, which is mixed with the relatively cold compression gas to enhance the temperature and pressure of the compression gas, wherein the mixed gas is fed from the gas generator to a cushion through a diffuser. In such a gas generator known in the art, the ignition tube is provided with a base, wherein a machined metal body is adopted to ensure a support strength of the base; alternatively, a metal body is adopted as a base main body, and a coating plastic body is applied on the base main body to form an integral structural unit by the ignition tube and the base in an injection molding process, in order to ensure the support strength of the base. Such an ignition system has a large size, a heavy weight and high manufacturing costs. Relevant patent literatures in the prior art may be referred to CN107000670B and CN111565980A.

SUMMARY OF INVENTION

[0007] An object of the present application is to provide an ignition system for a gas generator, a method for manufacturing the ignition system, a gas generator comprising the ignition system, and a safety airbag comprising the gas generator, wherein the ignition system may have a compact size, is lightweight, and may advantageously reduce the manufacturing costs.

[0008] A first aspect of the invention relates to an ignition system for a gas generator, which comprises an ignition tube, a base and a container. The base has a receptacle, wherein the ignition tube is press-fitted into the receptacle and extends out of the base, wherein the base is configured as an integral injection-molded part, wherein a metal inlay is injection-molded into a plastic material, and the inlay provides a portion of the receptacle. The container has an opening, through which the base together with the ignition tube is press-fitted into the container, with a head of the ignition tube facing forward.

[0009] Through the technical measures of the present invention, an improved gas generator, for example a hybrid gas generator, can be achieved. On one hand, disadvantages related to a welding process of a rupture membrane on a container in a manufacturing process of an ignition system in the prior art can be avoided. On the other hand, in comparison with the prior art in which a metal body is used as the base simply, or a metal body is used as a base main body and is coated with a plastic body thereon, the present invention can meet the requirements for overall lightweight of the ignition system, simplify the manufacturing process, and achieve a compact size, wherein the base may have a flexible mounting interface conveniently adapted to ignition tubes in different sizes and types.

[0010] In some embodiments, the inlay may be configured as a metal disc or comprise a metal disc, wherein the disc has a central receiving hole which provides the portion of the receptacle.

[0011] In some embodiments, the receiving hole may comprise a first cylindrical hole section adjacent to a first side of the disc, and a second tapered hole section adjacent to the first cylindrical hole section and widened towards a second side of the disc opposite to the first side, and the plastic material of the base defines a third cylindrical hole section adjacent to the second tapered hole section. The second tapered hole section may have the diameter of the first cylindrical hole section on one side thereof and the diameter of the third cylindrical hole section on the other side thereof.

[0012] In some embodiments, the inlay may have a structure for enhancing a bonding connection of the plastic material with the inlay.

[0013] In some embodiments, the structure may comprise a plurality of holes, such as blind holes or through holes, which are distributed around the central receiving hole, and/or a plurality of notches, which are distributed on an edge of the inlay.

[0014] In some embodiments, the structure may comprise a plurality of through holes, which are evenly distributed around the central receiving hole, and a plurality of notches, which are evenly distributed on the edge of the inlay, wherein in a circumferential direction with reference to an axis of the receiving hole, each of the plurality of notches is centrally arranged between respective two adjacent through holes of the plurality of through holes.

[0015] In some embodiments, the container may be necked at an opening side, so as to hold the base together with the ignition tube within the container.

[0016] In some embodiments, the base may comprise a first section, and a second section adjacent to the first section and having a reduced diameter in comparison with the first section, wherein the inlay is embedded in the plastic material of the first section.
[0017] In some embodiments, except for the metal disc, the remaining base is completely made of the plastic material.
[0018] In some embodiments, in addition to the metal disc, the inlay may further comprise a hollow cylindrical metal handle connected with the disc.
[0019] In some embodiments, in addition to the metal disc as the inlay, the base may further comprise an additional metal inlay separate from the metal disc.
[0020] In some embodiments, the base may comprise a first section, and a second section adjacent to the first section and having a reduced diameter in comparison with the first section, wherein the disc of the inlay is embedded in the plastic material of the first section, the handle of the inlay surrounds and defines the second section, wherein the plastic material is injection-molded on an inner peripheral surface of the handle of the inlay to form a cavity, wherein the plastic material in the first section and the plastic material in the second section of the base are integrally connected.
[0021] In some embodiments, the container may be necked at an opening side after the base together with the ignition tube is press-fitted into the container.
[0022] In some embodiments, the container may comprise a hollow body portion having the opening and a bottom opposite to the opening; and a hollow protrusion protruding from the bottom of the body portion.
[0023] In some embodiments, the base may be received in the body portion and is abutted against the bottom of the body portion, and the head of the ignition tube extending out of the base may be received in the protrusion, wherein a top end of the protrusion is configured to be opened by a shock wave generated by the ignition tube when the ignition system is activated.
[0024] In some embodiments, the protrusion may have a wall thickness, which is gradually decreased in a direction away from the body portion, and/or the protrusion is provided with a weakened notch at its top end. With the design of a gradually-varying wall thickness for the hollow protrusion of the container, and/or with a weakened design for a thin-wall area on the top end of the protrusion, the shock wave generated by the ignition tube, when the ignition tube is activated, can smoothly open the top end of the protrusion of the container, so as to reliably ignite the pyrotechnic charge contained in the pyrotechnic charge chamber.
[0025] In some embodiments, a ratio “a” of a minimum wall thickness to a maximum wall thickness of the protrusion can satisfy the formula $0.5<a<1$, and/or a ratio “b” of a height of the protrusion to an outer diameter of the head of the ignition tube can satisfy the formula $b\geq 1$. Under the circumstance that the hollow protrusion of the container has the predetermined height, a reliable ignition channel can be provided to ensure the operation reliability of igniting the pyrotechnic charge in the pyrotechnic charge chamber of the gas generator, especially the hybrid gas generator.
[0026] In some embodiments, the ignition system may be a rotationally symmetric structural unit with a longitudinal axis.
[0027] In some embodiments, the base and/or the container and/or the ignition tube may be rotationally symmetric members with a longitudinal axis.
[0028] In some embodiments, the base comprises a front end surface and a rear end surface. Preferably, at least one, and particularly both, of the end surfaces is/are planar.
[0029] In some embodiments, the container may be internally provided, in a transition area from the body portion to the protrusion, with a slope adapted to the ignition tube, so as to provide a supporting and fixing interface for the ignition tube.
[0030] A second aspect of the invention relates to a gas generator, which comprises an ignition system according to any one of embodiments of the present invention, and a pyrotechnic charge chamber, wherein the ignition system and the pyrotechnic charge chamber are configured such that the ignition system, when activated, ignites a pyrotechnic charge contained in the pyrotechnic

charge chamber.
[0031] In some embodiments, the gas generator may be a hybrid gas generator. The hybrid gas generator comprises a pressure container, in which a pressure chamber for containing the compression gas is defined, wherein the ignition system and the pyrotechnic charge chamber are configured such that the ignition system, when activated, ignites the pyrotechnic charge contained in the pyrotechnic charge chamber, wherein the gas generated by the combustion of the pyrotechnic charge can be mixed with the compression gas.
[0032] In some embodiments, the gas generator may be configured for a driver's safety airbag, a co-driver's safety airbag, a side air curtain, a seat safety airbag or any other safety airbag.
[0033] A third aspect of the invention relates to a method for manufacturing an ignition system, comprising the steps of: [0034] providing an ignition tube; [0035] providing a base having a receptacle and configured as an integral injection-molded part, wherein a metal inlay is injection-molded into a plastic material, and the inlay provides a portion of the receptacle; [0036] providing a container having an opening; [0037] press-fitting the ignition tube into the receptacle, wherein the ignition tube extends out of the base with a head of the ignition tube; [0038] press-fitting the base together with the ignition tube into the container through the opening, with the head of the ignition tube facing forward.
[0039] In some embodiments, the method may further comprise necking the container at the opening side, after the base together with the ignition tube is press-fitted into the container.
[0040] A fourth aspect of the invention relates to a safety airbag, which comprises a cushion, in particular a folded cushion, and a gas generator according to any one of embodiments of the present invention, wherein the safety airbag is preferably a side air curtain.
[0041] A fifth aspect of the present invention relates to a transportation means, in particular a motor vehicle, comprising the safety airbag according to the fourth aspect of the invention.
[0042] The technical features mentioned above, technical features to be mentioned below and technical features separately shown in the drawings can be arbitrarily combined with each other, only if the combined technical features are not contradictory. All the feasible combinations of the features belong to the technical contents clearly recited in the present application. Any one of the multiple sub-features contained in a single sentence can be applied independently, and need not to be applied together with the other sub-features.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0043] The present invention is described in more detail by means of exemplary embodiments with reference to the attached drawings. The figures are briefly described as follows:
[0044] FIG. **1** is a schematic longitudinal sectional view of a gas generator according to an embodiment of the present invention.
[0045] FIG. **2** is a schematic longitudinal sectional view of an ignition system of the gas generator of FIG. **1**.
[0046] FIGS. **3**A and **3**B are a perspective view and a longitudinal sectional view of a base of the ignition system of FIG. **2**.
[0047] FIG. **3**C is a perspective view of a metal inlay of the base of FIGS. **3**A and **3**B.
[0048] FIG. **3**D is a schematic longitudinal sectional view of a container before being assembled.
[0049] FIGS. **4**A to **4**C are schematic top views of a protrusion of the container of FIG. **3**D.
[0050] FIGS. **5**A to **5**C are schematic views showing steps for manufacturing the ignition system of FIG. **2**.
[0051] FIGS. **6**A and **6**B are a perspective view and a longitudinal sectional view of a base according to another embodiment.

[0052] FIG. **6**C is a perspective view of a metal inlay of the base of FIGS. **6**A and **6**B.

EMBODIMENTS

[0053] Exemplary embodiments are now described more fully with reference to the attached drawings. It should be noted that elements that are not necessary for understanding the present invention may be omitted from the drawings for convenience of explanation and understanding. In the drawings, the same reference signs indicate the same components or components with the same function. In the following description, numerous specific details are set forth, such as examples of specific components, devices and methods, in order to provide a thorough understanding of embodiments of the disclosure. It will be apparent to those skilled in the art that it is not necessary to adopt all these specific details. The exemplary embodiments should not be understood as restrictive ones.

[0054] FIG. **1** is a schematic longitudinal sectional view of a gas generator **100** according to an embodiment of the present invention. The gas generator **100** may be configured as a hybrid gas generator. Although the present invention is introduced in particular with reference to a hybrid gas generator hereafter, those skilled in the art can, under the teaching provided herein, appreciate and recognize that the present invention is not limited to the hybrid gas generator, and the gas generator **100** of the present invention can be applied generally to various types of safety airbags, such as a knee safety airbag, a distal airbag, a seat-side airbag and the like. Those skilled in the art can appreciate and recognize that the ignition system **10**, the gas generator **100** comprising the ignition system **10**, and the safety airbag comprising the gas generator **100** can be generally applied to transportation means, such as land transportation means, marine transportation means or airborne transportation means, and particularly motor vehicles. The motor vehicles may relate to passenger vehicles or trucks.

[0055] The gas generator **100** shown in FIG. **1** may comprise an elongated pressure container **80** that extends in a straight or curved line, such as a pressure container **80** in the shape of a hollow cylinder, in which a pressure chamber **30** for containing the compression gas is formed. The compression gas may typically be an inert gas. The pressure container **80** may be constructed as a tubular member, which may have a circular or non-circular cross section. The pressure container **80** is provided with an ignition system **10** and a pyrotechnic charge chamber **20** at its first axial end, and with a diffuser **40** at its second axial end opposite to the first axial end. The pressure chamber **30** is hermetically closed by the ignition system **10** at the first axial end, and is hermetically closed by the diffuser **40** at the second axial end. The ignition system **10** is, as a structural unit, fixed to the first axial end of the pressure container **80**. For example, it's hermetically connected with an inner wall of the pressure container **80** by welding. The pyrotechnic charge chamber **20** is defined by a first partition element **50** and a second partition element **60**, and is filled with a pyrotechnic charge in the form of block, sheet, ball or any other suitable forms. The first partition element **50** is abutted against the ignition system **10**, and has a central hole, into which a protrusion **32** of the ignition system **10** extends. The second partition element **60** may have a plurality of small holes, which shall be sized to ensure that the pyrotechnic charge can't pass from the pyrotechnic charge chamber **20** through these small holes into the pressure chamber **30** containing the compressed air. The diffuser **40** may be sealed with a rupture membrane **70** on its end surface facing the pressure container **80**.

[0056] When the ignition system **10** is activated, the pyrotechnic charge contained in the pyrotechnic charge chamber **20** is ignited by the ignition system **10**, wherein a hot gas with a high temperature and a high pressure is generated by combustion of the pyrotechnic charge. The hot gas enters the pressure chamber **30** through the small holes in the second partition element **60**, and is mixed with the cold compression gas to enhance the temperature and pressure of the cold compression gas. Accordingly, the rupture membrane **70** on the diffuser **40** is broken under the action of a shock wave and/or the pressure, and then the mixed gas is output to a folded cushion (not shown) of the safety airbag through the diffuser **40**, inflates the cushion, and thereby a

predetermined safety protection function is provided.
[0057] In the embodiment shown in FIG. **1**, the ignition system **10** is applied to a hybrid gas generator. It can be understood that the ignition system **10** is not limited to be applied in a hybrid gas generator, but can be applied to various types of gas generators.
[0058] The ignition system **10** for the gas generator **100** and components of the ignition system **10** are now described in more detail with reference to FIG. **2**, FIGS. **3**A to **3**D and FIGS. **4**A to **4**C.
[0059] The ignition system **10** comprises an ignition tube **1**, a base **2** and a container **3**. The ignition tube **1** itself is known in practice and may be a standardized product. The ignition tube **1** may also be referred to as a detonator or a detonator assembly. The ignition tube **1** may contain reactive fillers. When an electrically conductive lead wire **12** of the ignition tube **1** is powered on, and then chemical reaction occurs to the reactive fillers rapidly, wherein a shock wave is generated to open a top end of the container **3**, and thus the pyrotechnic charge in the pyrotechnic charge chamber **20** is ignited to fulfill the detonation function.
[0060] The base **2** is configured as an integral injection-molded part, wherein a metal inlay **4** is injection-molded into a plastic material. The plastic material may be a thermoplastic or thermosetting plastic. The plastic material may be a single plastic or a combination of multiple plastics. The plastic material may contain reinforcing fibers, such as glass fibers or carbon fibers. The plastic material may be an elastomeric plastic or a plastic with substantially no elasticity. Exemplary plastic materials may be F30, F40, PA, PC, PVC and PPR. The metal material for the inlay **4** may be, for example, carbon steel and stainless steel. The material for the container **3** may be the same as or different from the plastic material for the base **2**, for example, may be glass fiber- or carbon fiber-reinforced plastic.
[0061] As shown in FIGS. **3**A and **3**B, the base **2** comprises a first section **21** with a larger diameter, and a second section **22** adjacent to the first section **21** and having a reduced diameter in comparison with the first section **21**. The inlay **4** is embedded in the plastic material of the first section **21**. The first section **21** and the second section **22** may have an arc-shaped transition **26** therebetween. The base **2** has a front end surface **24** and a rear end surface **25**. The base **2** has a receptacle **23** open to the front end surface **24**. The ignition tube **1** can be press-fitted into the receptacle **23** of the base **2**. In a state where the ignition tube **1** and the base **2** are assembled, the ignition tube **1** extends out of the front end surface **24** of the base **2**.
[0062] The receptacle **23** of the base **2** comprises a plurality of hole sections **231**, **232** and **233**. As shown in FIGS. **3**B and **3**C, the inlay **4** may be constructed as a metal disc. The inlay **4** has a central receiving hole **41**. The receiving hole **41** comprises a first cylindrical hole section **231** adjacent to a first side of the inlay **4**, and a second tapered hole section **232** adjacent to the first cylindrical hole section **231** and widened towards a second side of the inlay **4** opposite to the first side. The inlay **4** provides the two hole sections **231**, **232** of the receptacle **23** of the base **2**. The plastic material of the first section **21** of the base **2** defines a third cylindrical hole section **233** adjacent to the second tapered hole section **232**, wherein the third cylindrical hole section is a portion of the receptacle **23**. The plastic material of the base **2** in the second section **22** defines a recess **234**, into which the lead wire **12** of the ignition tube **1** extends. Advantageously, the base **2** may be a rotationally symmetrical body.
[0063] In order to enhance a bonding strength between the plastic material and the metal inlay **4** of the base **2**, the inlay **4** may have a structure. The structure may be a protrusion such as a pillar, a bump, a pyramid or the like, which extends from the surface of the inlay. Alternatively or additionally, the structure may be a recess, such as a through hole, a blind hole, a notch or the like. In the exemplary embodiment shown in FIG. **3**C, the structure comprises six through holes **42**, which are evenly distributed around the receiving hole **41**, and six notches **43**, which are evenly distributed on an edge of the inlay **4**, wherein in a circumferential direction with reference to an axis of the receiving hole **41** of the inlay **4**, each of the six notches **43** is centrally arranged between respective two adjacent through holes of the six through holes **42**, with an angular distance of 60°

between two adjacent through holes **42**, with an angular distance of 60° between two adjacent notches **43**, and with an angular distance of 30° between each notch **43** and any adjacent through hole **42**.

[0064] As shown in FIG. **2** and FIG. **3**D, the container **3** comprises a hollow body portion **31**, which has an opening **33** and a bottom **37** opposite to the opening **33**. The container **3** further comprises a hollow protrusion **32** protruding from the bottom **37** of the body portion **31**. The protrusion **32** may comprise a short hollow cylindrical portion and a hollow hemispherical portion. As shown in FIG. **3**D, the body portion **31** of the container **3** is cylindrical before being assembled. During assembly, the base **2** together with the ignition tube **1**, is press-fitted into the container **3** through the opening **33**, with a head **11** of the ignition tube **1** facing forward (i.e., facing the container).

[0065] As shown in FIG. **2**, after being assembled, the container **3** is necked at an axial section adjacent to the opening **33**, thus forming a first section **34** with a larger diameter and a second section **35** with a reduced diameter. With necking of the container **3**, the base **2** together with the ignition tube **1** is firmly held in the container **3**. Even when the ignition system **10** is activated, the base **2** together with the ignition tube **1** isn't detachable from the container **3** through the opening **33** of the container **3**.

[0066] In an assembled state of the ignition system **10**, the base **2** is received in the body portion **31** and is abutted against the bottom **37** of the body portion **31**, and the head **11** of the ignition tube **1** extending out of the base **2** is received in the protrusion **32**. A top end of the protrusion **32** is configured to be openable by a shock wave generated by the ignition tube **1** when the ignition system **10** is activated. The first section **34** of the body portion **31** of the container **3** surrounds the first section **21** of the base **2**, and the second section **35** of the body portion **31** of the container **3** surrounds the second section **22** of the base **2**. The container **3** may be internally provided, in a transition area from the body portion **31** to the protrusion **32**, with a slope **36** adapted to the ignition tube **1**, so as to provide a supporting and fixing interface for the ignition tube **1**.

[0067] Advantageously, the protrusion **32** has a wall thickness, which is gradually decreased in a direction away from the body portion **31**, and thus has a minimum wall thickness at the top end and a maximum wall thickness at the root adjacent to the bottom **37** of the body portion **31**. Advantageously, a ratio "a" of a minimum wall thickness to a maximum wall thickness of the protrusion **32** may satisfy the formula: $0.5<a<1$. Furthermore, it is advantageous that a ratio "b" of a height H of the protrusion to an outer diameter D of the head **11** of the ignition tube **1** may satisfy the formula: $b\geq1$.

[0068] The protrusion **32** may have a weakened notch at its top end or not. The weakened notch may be in the form of a straight line, a cross, a plum blossom, a star or any other suitable form. FIGS. **4**A-**4**C show three exemplary weakened notches. In the example shown in FIG. **4**A, the top end of the protrusion **32** is provided with three intersecting notches in the form of straight lines. In the example shown in FIG. **4**B, the top end of the protrusion **32** is provided with a notch in the form of a plum blossom. In the example shown in FIG. **4**C, the top end of the protrusion **32** is provided with a notch in the form of a five-pointed star.

[0069] The container **3** may have a gradually varying wall thickness, wherein a larger wall thickness is adopted for the hollow body portion **31** so as to ensure, particularly well, the assembly of the container **3** with other components of the gas generator **100** and guarantee a sufficient fastening strength of the container **3** for the base at its opening side; and a gradually-decreasing wall thickness is adopted for the hollow protrusion **32**, preferably the height H of the protrusion being greater than or equal to the outer diameter D of the head **11** of the ignition tube **1**, so that the shock wave generated by the ignition tube **1** can open the closed top end of the protrusion **32** timely when the ignition tube **1** is activated.

[0070] FIGS. **5**A to **5**C are schematic views showing various steps for manufacturing the ignition system **100** shown in FIG. **2**.

[0071] As shown in FIG. **5**A, after the ignition tube **1** and the base **2** are provided, the ignition tube **1** is press-fitted into the receptacle **23** of the base **2**, as schematically indicated by the arrow P**1**. The ignition tube **1** and the receptacle **23** of the base **2** have complementary shapes, wherein the ignition tube **1** is firmly held in the receptacle **23** of the base **2** in an interference-fitting manner, and the two electrically-conductive lead wires **12** of the ignition tube **1** extend into the recess **234** at the rear side of the base **2**. Upon press-fitting, the base **2** may be kept still, while the ignition tube **1** is moved; or the ignition tube **1** is kept still, while the base **2** is moved; or not only the base **2** but also the ignition tube **1** are moved.
[0072] As shown in FIG. **5**B, after the base **2** is assembled with the ignition tube **1**, the container **3** is provided, and the base **2** together with the ignition tube **1** is press-fitted, with the head **11** of the ignition tube **1** facing forward, into the container **3** through the opening **33** of the container **3**, as schematically indicated by two arrows P**2**. Upon press-fitting, the assembled base **2** and ignition tube **1** may be kept still, while the container **3** is moved; or the container **3** is kept still, while the assembled base **2** and ignition tube **1** are moved; or not only the assembled base **2** and ignition tube **1**, but also the container **3** are moved.
[0073] As shown in FIG. **5**C, after the assembled base **2** and ignition tube **1** are press-fitted into the container **3**, the container **3** is necked at the opening side. The necking step may be realized, for example, by a riveting device **90** schematically illustrated. As an alternative or supplement, the necking may be realized by an extrusion tool.
[0074] Now, a base **2** according to another embodiment is described with reference to FIGS. **6**C to **6**C. The base **2** may be assembled with an ignition tube **1** and a container **3**, which may be constructed in a way same as or similar to those in the ignition system **10** shown in FIG. **2**, to form an ignition system according to another embodiment, wherein the assembling method may be carried out in the same or similar way as that shown in FIGS. **5**A to **5**C. The base **2** is configured as an integral injection-molded part, in which a metal inlay **4** is injection-molded into a plastic material.
[0075] As shown in FIG. **6**C, the metal inlay **4** comprises a metal disc, and a hollow cylindrical metal handle **45** connected, especially integrally connected, to the disc. Here, the disc may be constructed in a way same as or similar to that shown in FIG. **3**C.
[0076] As shown in FIGS. **6**A and **6**B, the base **2** comprises a first section **21** with a larger diameter, and a second section **22** adjacent to the first section **21** and having a reduced diameter in comparison with the first section **21**. The metal disc of the inlay **4** is embedded in the first plastic material **48** of the first section **21**, the handle **45** of the inlay **4** surrounds and defines the second section **22**, the second plastic material **47** is injection-molded on an inner peripheral surface of the handle **45** of the inlay **4** and thus forms a cavity, wherein the first plastic material **48** in the first section and the second plastic material **47** in the second section of the base **2** are integrally connected.
[0077] Same as or Similar to the previous embodiment, the base **2** has a front end surface **24** and a rear end surface **25**. The base **2** has a receptacle **23** open to the front end surface **24**. An ignition tube (not shown) can be press-fitted into the receptacle **23** of the base **2**. The front end surface **24** of the base **2** is provided by the first plastic material **48**. The rear end surface **25** of the base **2** is provided jointly by the handle **45** of the metal inlay **4** and the second plastic material **47**.
[0078] As shown in FIGS. **6**A and **6**B, the through holes **42** in the disc of the inlay **4** are filled with the plastic material during injection molding, so that the injection molded material and the metal inlay are firmly connected together. Unlike the embodiment shown in FIGS. **3**A and **3**B, here, the notches **43** on the edge of the disc of the inlay **4** are not used for the interconnection between the injection molded material and the metal inlay. Instead, the notches **43** together with the corresponding notches in the first plastic material **48** form notches **46** running through the entire axial length of the first section.
[0079] The base **2** as shown in FIGS. **6**A and **6**B can be assembled with the container **3** by rotary

riveting particularly well. The base **2** may have improved overall strength, and thus have the advantages of light weight and high strength.

[0080] It should be noted that within the scope of the present application, the cylindrical shape may preferably be a circular cylindrical shape or a substantially circular cylindrical shape. It may be appreciated that other cylindrical shapes or approximately cylindrical shapes that deviate from the circular cylindrical shape may also be taken into consideration.

[0081] It should be noted that the terminology used here is only for the purpose of describing specific aspects, and is not used to limit the disclosure. As used herein, the singular forms “a” and “the” shall comprise the plural forms, unless otherwise explicitly indicated. It can be understood that the terms “comprising” and “including” and other similar terms, when used in the application document, specify the existence of stated operations, elements and/or components, but do not exclude the existence or addition of one or more other operations, elements, components and/or their combinations. The term “and/or” as used herein comprises all arbitrary combinations of one or more related listed items. In the description of the drawings, like reference signs always indicate like elements.

[0082] The thickness of elements in the drawings may be exaggerated for clarity. In addition, it can be appreciated that if an element is referred to be on, coupled with or connected with another element, it can be directly formed on, coupled with or connected with the other element, or there may be one or more intervening elements between them. On the contrary, if the expressions “directly on . . . ”, “directly coupled with . . . ” and “directly connected with . . . ” are used here, it means that there are no intervening elements. Other wordings such as “between . . . ” and “directly between . . . ”, “attached” and “directly attached”, “adjacent” and “directly adjacent” and so on used to describe the relationship between elements should be interpreted in a similar way.

[0083] Terms such as “top”, “bottom”, “above”, “below”, “upper” and “lower” are used here to describe the relationship of one element, layer or region with respect to another element, layer or region as shown in the drawings. It can be appreciated that these terms should also comprise other orientations of the device besides those depicted in the drawings.

[0084] It can be understood that although the terms “first”, “second” and the like may be used herein to describe different elements, these elements shall not be limited by these terms. These terms are only used to distinguish one element from another. Therefore, a first element may be referred to as a second element without departing from the teaching of the present inventive concept.

[0085] It can also be contemplated that all the exemplary embodiments disclosed herein can be arbitrarily combined with each other. Finally, it should be pointed out that the above embodiments are only for understanding the present invention, and are not to limit the scope of protection of the present invention. For those skilled in the art, modifications can be made on the basis of the embodiments, which do not depart from the protection scope of the present invention.

Claims

1. An ignition system for a gas generator, comprising an ignition tube, wherein the ignition system further comprises: a base having a receptacle, wherein the ignition tube is press-fitted into the receptacle and extends out of the base with its head, wherein the base is configured as an integral injection-molded part, wherein a metal inlay is injection-molded into a plastic material, wherein the inlay provides a portion of the receptacle; and a container having an opening, through which the base together with the ignition tube is press-fitted into the container, with the head of the ignition tube facing forward.

2. The ignition system for a gas generator as recited in claim 1, wherein the inlay is configured as a metal disc or comprises a metal disc, wherein the disc has a central receiving hole [(41)] which provides the portion of the receptacle.

3. The ignition system for a gas generator as recited in claim 2, wherein the receiving hole comprises a first cylindrical hole section adjacent to a first side of the disc, and a second tapered hole section adjacent to the first cylindrical hole section and widened towards a second side of the disc opposite to the first side, and the plastic material of the base defines a third cylindrical hole section adjacent to the second tapered hole section.
4. The ignition system for a gas generator as recited in claim 2, wherein the disc has a structure for enhancing a bonding connection of the plastic material with the inlay.
5. The ignition system for a gas generator as recited in claim 4, wherein the structure comprises a plurality of through holes distributed around the central receiving hole, and/or a plurality of notches distributed on an edge of the inlay.
6. The ignition system for a gas generator as recited in claim 4, wherein the structure comprises a plurality of through holes, which are evenly distributed around the central receiving hole, and a plurality of notches, which are evenly distributed on an edge of the inlay, wherein in a circumferential direction with reference to an axis of the receiving hole, each of the plurality of notches is centrally arranged between respective two adjacent through holes of the plurality of through holes.
7. The ignition system for a gas generator as recited in claim 2, wherein in addition to the metal disc, the inlay further comprises a hollow cylindrical metal handle connected with the disc.
8. The ignition system for a gas generator as recited in claim 1, wherein the container is necked at the opening side to hold the base together with the ignition tube within the container.
9. The ignition system for a gas generator as recited in any claim 1, wherein the base comprises a first section, and a second section adjacent to the first section and having a reduced diameter in comparison with the first section, wherein the inlay is embedded in the plastic material of the first section, and the container is necked at the opening side after the base together with the ignition tube is press-fitted into the container.
10. The ignition system for a gas generator as recited in claim 7, wherein the base comprises a first section, and a second section adjacent to the first section and having a reduced diameter in comparison with the first section, wherein the disc of the inlay is embedded in the plastic material of the first section, the handle of the inlay surrounds and defines the second section, and the plastic material is injection-molded on an inner peripheral surface of the handle of the inlay and thus forms a cavity, wherein the plastic material in the first section and the plastic material in the second section of the base are integrally connected.
11. The ignition system for a gas generator as recited in claim 1, wherein the container comprises: a hollow body portion having the opening and a bottom opposite to the opening; and a hollow protrusion protruding from the bottom of the body portion; wherein the base is received in the body portion and is abutted against the bottom of the body portion, wherein the head of the ignition tube extending out of the base is received in the protrusion, and a top end of the protrusion is configured to be opened by a shock wave generated by the ignition tube when the ignition system is activated.
12. The ignition system for a gas generator as recited in claim 11, wherein the protrusion has a wall thickness, which is gradually decreased in a direction away from the body portion, and/or the protrusion has a weakened notch at its top end.
13. The ignition system for a gas generator as recited in claim 12, wherein a ratio “a” of a minimum wall thickness to a maximum wall thickness of the protrusion satisfies the formula $0.5<a<1$, and/or a ratio “b” of a height of the protrusion to an outer diameter of the head of the ignition tube satisfies the formula $b\geq1$.
14. The ignition system for a gas generator as recited in claim 11, wherein the container is internally provided, in a transition area from the body portion to the protrusion, with a slope adapted to the ignition tube, so as to provide a supporting and fixing interface for the ignition tube.
15. A gas generator comprising an ignition system and a pyrotechnic charge chamber, wherein the ignition system and the pyrotechnic charge chamber are configured such that the ignition system,

when activated, ignites a pyrotechnic charge contained in the pyrotechnic charge chamber, wherein the ignition system is the one as recited in claim 1.

16. The gas generator as recited in claim 15, wherein the gas generator is a hybrid gas generator, wherein the hybrid gas generator comprises a pressure container in which a pressure chamber for containing a compression gas is defined, wherein the ignition system and the pyrotechnic charge chamber are configured such that the ignition system, when activated, ignites the pyrotechnic charge contained in the pyrotechnic charge chamber, and gas generated by combustion of the pyrotechnic charge is mixable with the compression gas.

17. A safety airbag comprising a cushion, and a gas generator in a fluid connection with the cushion, wherein the gas generator is the one as recited in claim 15.

18. A method for manufacturing the ignition system as recited in claim 1, the method comprising the steps of: providing an ignition tube; providing a base having a receptacle and configured as an integral injection-molded part, wherein a metal inlay is injection-molded into a plastic material, wherein the inlay provides a portion of the receptacle; providing a container having an opening; press-fitting the ignition tube into the receptacle, wherein the ignition tube extends out of the base with its head; press-fitting the base together with the ignition tube into the container through the opening, with the head of the ignition tube facing forward.

19. The method as recited in claim 18, wherein the base comprises a first section, and a second section adjacent to the first section and having a reduced diameter in comparison with the first section, the disc being embedded in a plastic material of the first section; wherein the method comprises: necking the container at the opening side after the base together with the ignition tube is press-fitted into the container.